

Product Purchase Data Analysis using Unix Command-Line Tools

1. Introduction

In modern organizations, a large amount of transactional data is generated every day. This data is often stored in log files or simple text files before being processed by large data processing systems. Unix command-line tools provide powerful utilities that allow quick analysis of such textual datasets without requiring complex programming. These tools help in filtering, sorting, counting, and extracting useful information from structured text files.

The purpose of this activity was to perform basic data exploration and analysis using Unix command-line tools on a dataset containing product purchase records. The dataset consisted of multiple purchase transactions, each containing information about the transaction ID, product name, and product category. Using Unix utilities, various analytical tasks such as counting total records, extracting specific columns, identifying category-wise purchase distribution, and determining frequently purchased products were performed.

2. Dataset Preparation

The first step in this activity was creating a dataset file named **purchases.txt** inside a working directory. The dataset contained 50 purchase records. Each record followed a structured format with three fields:

TransactionID ProductName Category

Example records from the dataset:

T001 Laptop Electronics
T002 Mobile Electronics
T003 Shirt Clothing
T004 Shoes Footwear
T005 Laptop Electronics

Each field was separated by a space delimiter. This structure made it easier to extract individual columns using Unix tools such as the **cut** command. The dataset included products from multiple categories such as Electronics, Clothing, Footwear, and Accessories.

3. Preprocessing Steps

Before performing analysis, the dataset was examined to ensure that it followed the correct structure. Several Unix commands were used for preprocessing and verification.

The **wc -l** command was used to count the total number of records present in the dataset. This confirmed that the dataset contained 50 purchase entries.

The **head** command was used to display the first few lines of the file to verify the format and structure of the dataset.

The **cut** command was used to extract specific columns from the dataset. Since the fields were separated by spaces, the delimiter option (`-d " "`) was used to separate columns.

These preprocessing steps ensured that the dataset was clean, structured, and ready for further analysis.

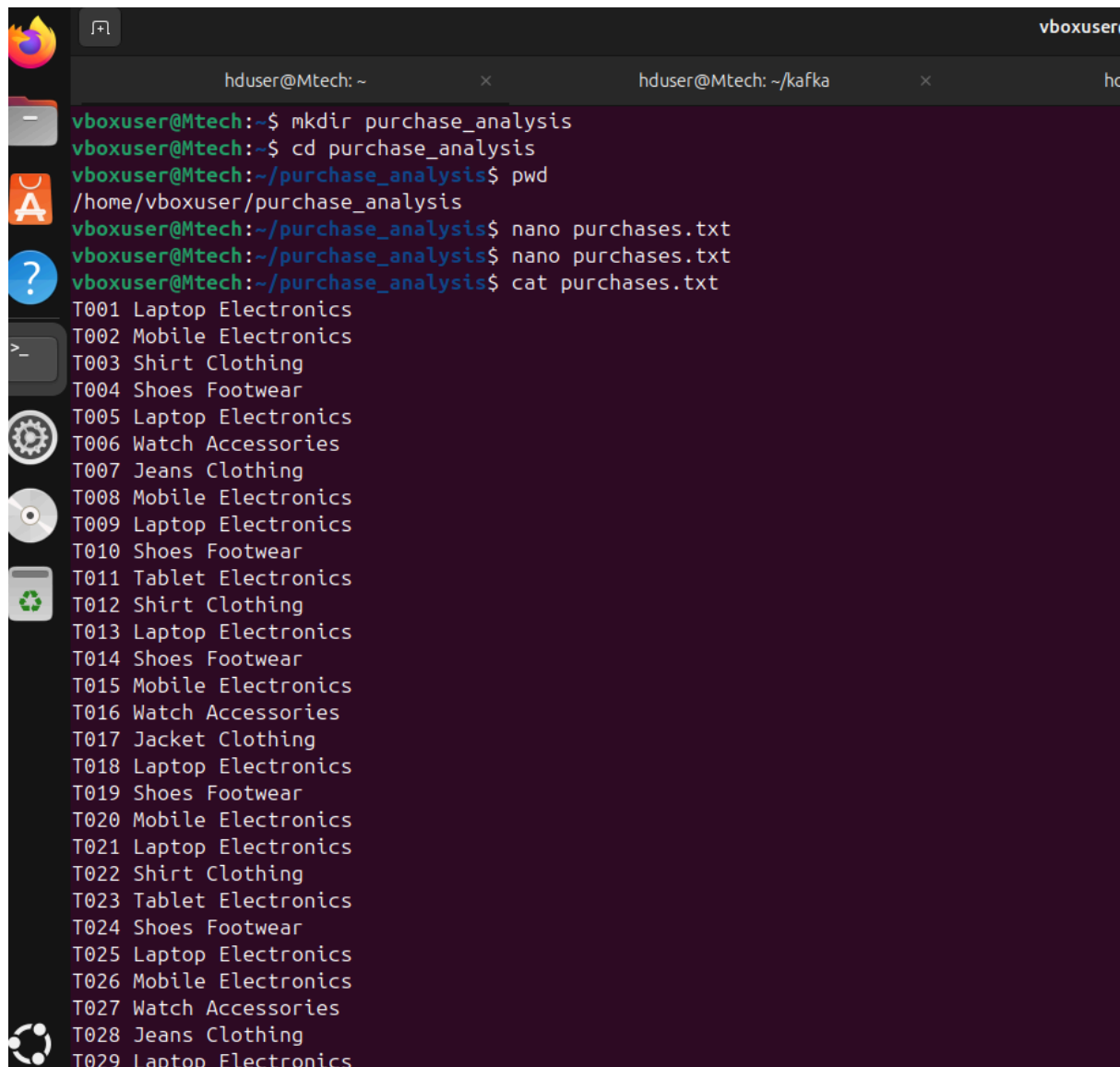
4. Data Analysis Using Unix Commands

4.1 Basic Dataset Exploration

Initial exploration was performed to understand the structure of the dataset. The following tasks were completed:

- Counting the total number of purchase records using `wc -l`
- Displaying the first few records using `head`
- Extracting the product column using the `cut` command

These commands helped verify that the dataset was correctly formatted and contained the required information.

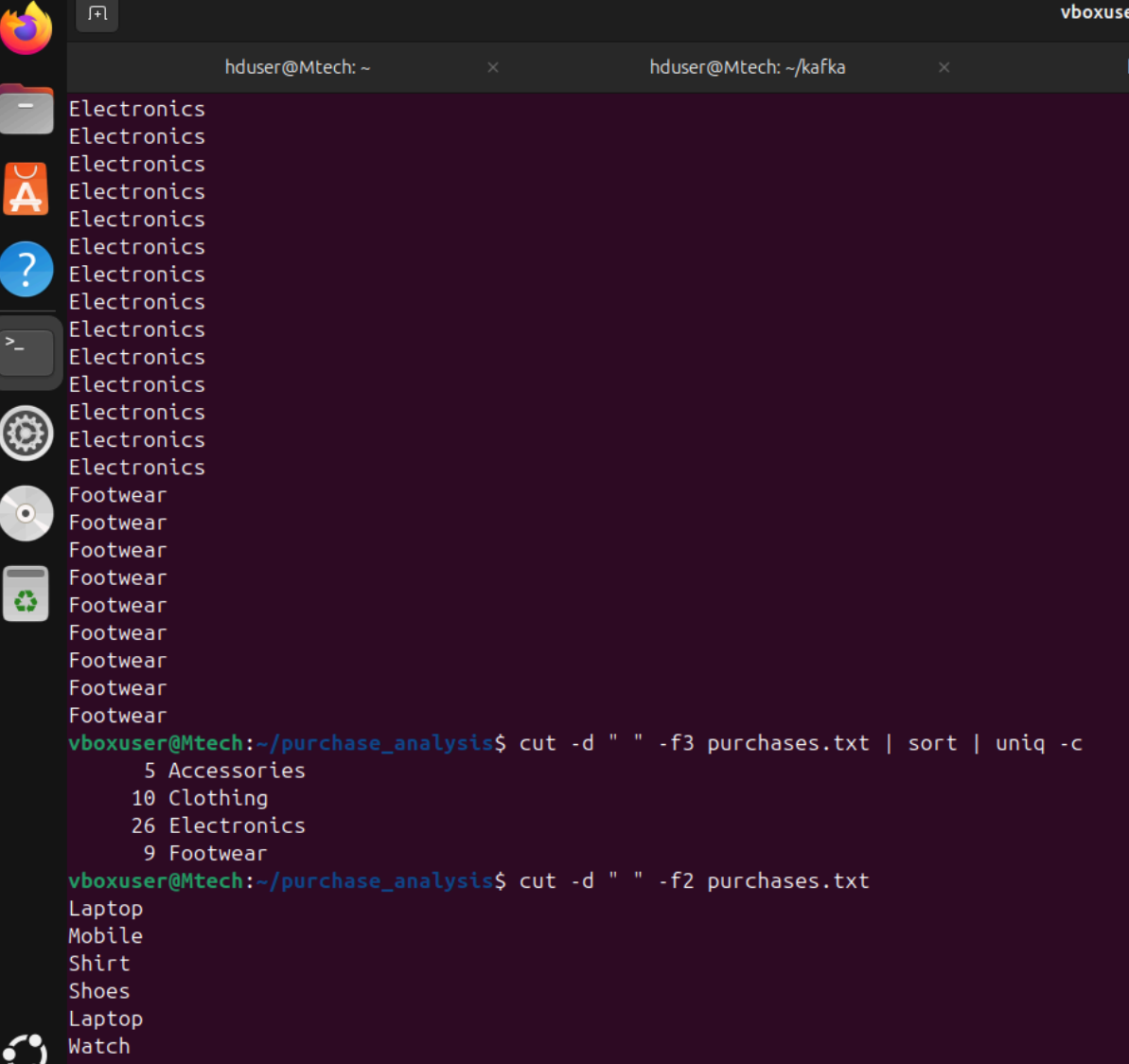


```
vboxuser@Mtech:~$ mkdir purchase_analysis
vboxuser@Mtech:~$ cd purchase_analysis
vboxuser@Mtech:~/purchase_analysis$ pwd
/home/vboxuser/purchase_analysis
vboxuser@Mtech:~/purchase_analysis$ nano purchases.txt
vboxuser@Mtech:~/purchase_analysis$ nano purchases.txt
vboxuser@Mtech:~/purchase_analysis$ cat purchases.txt
T001 Laptop Electronics
T002 Mobile Electronics
T003 Shirt Clothing
T004 Shoes Footwear
T005 Laptop Electronics
T006 Watch Accessories
T007 Jeans Clothing
T008 Mobile Electronics
T009 Laptop Electronics
T010 Shoes Footwear
T011 Tablet Electronics
T012 Shirt Clothing
T013 Laptop Electronics
T014 Shoes Footwear
T015 Mobile Electronics
T016 Watch Accessories
T017 Jacket Clothing
T018 Laptop Electronics
T019 Shoes Footwear
T020 Mobile Electronics
T021 Laptop Electronics
T022 Shirt Clothing
T023 Tablet Electronics
T024 Shoes Footwear
T025 Laptop Electronics
T026 Mobile Electronics
T027 Watch Accessories
T028 Jeans Clothing
T029 Laptop Electronics
```

4.2 Category-wise Purchase Analysis

To analyze purchases based on categories, the third column (Category) was extracted using the **cut** command. The extracted values were then sorted using the **sort** command. Sorting is important before counting occurrences because it groups identical values together.

After sorting, the **uniq -c** command was used to count how many purchases belonged to each category.

A terminal window with a dark purple background. The left sidebar shows various application icons. The terminal has two tabs: 'hduser@Mtech: ~' and 'hduser@Mtech: ~/kafka'. The first tab is active. The terminal shows a list of purchase categories: Electronics (repeated 10 times), Footwear (repeated 6 times), and a few others. Below this, the command 'cut -d " " -f3 purchases.txt | sort | uniq -c' is executed, resulting in: 5 Accessories, 10 Clothing, 26 Electronics, and 9 Footwear. Then, the command 'cut -d " " -f2 purchases.txt' is executed, resulting in a list of product names: Laptop, Mobile, Shirt, Shoes, Laptop, and Watch.

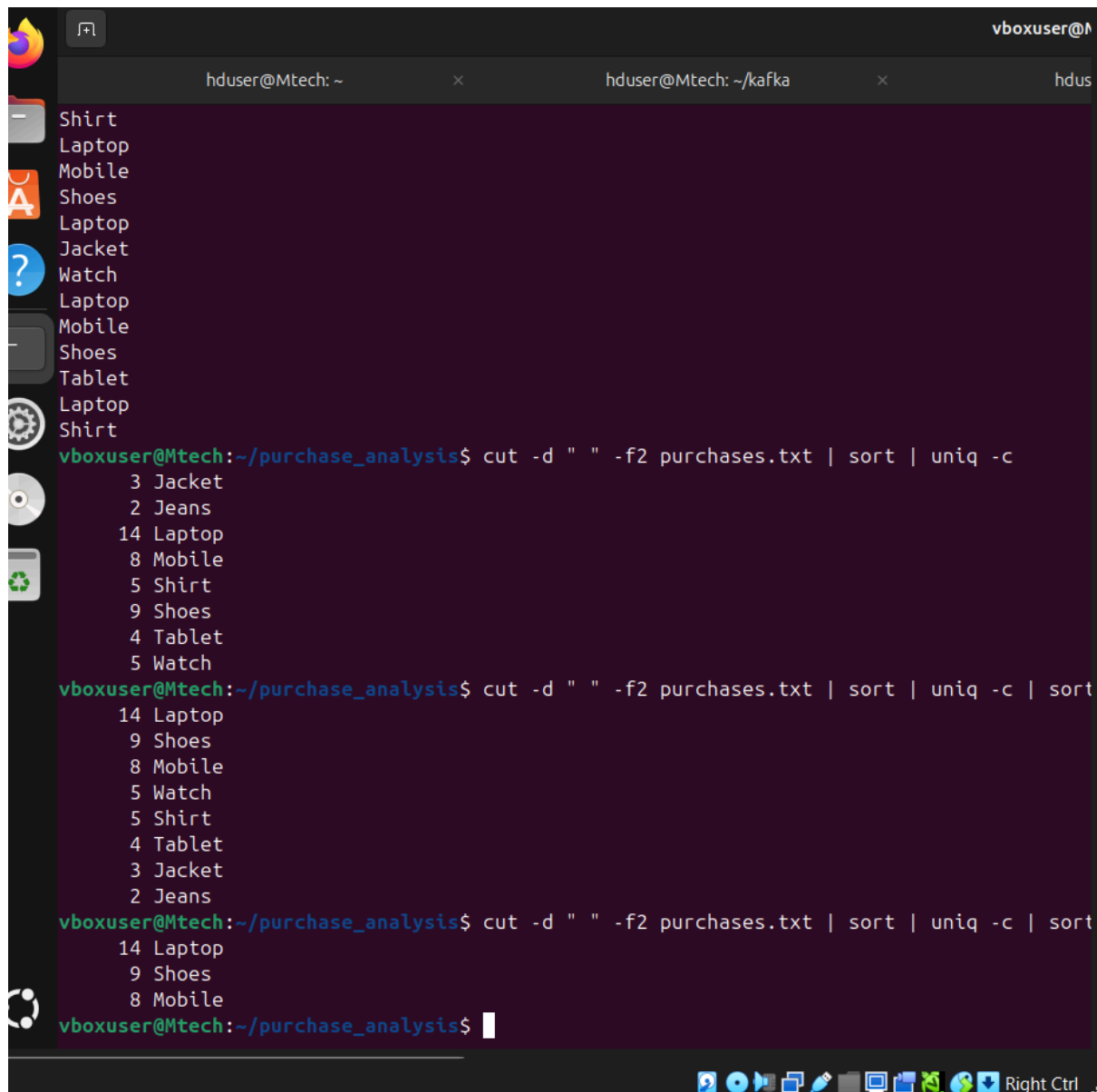
```
hduser@Mtech: ~  
hduser@Mtech: ~/kafka  
vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f3 purchases.txt | sort | uniq -c  
5 Accessories  
10 Clothing  
26 Electronics  
9 Footwear  
vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt  
Laptop  
Mobile  
Shirt  
Shoes  
Laptop  
Watch
```

This analysis revealed that the Electronics category had the highest number of purchases.

4.3 Most Frequently Purchased Products

To identify the most frequently purchased products, the second column (Product Name) was extracted from the dataset. The extracted product names were then processed using a pipeline of Unix commands.

First, the product column was sorted using the **sort** command. After sorting, the **uniq -c** command counted the occurrences of each product. Finally, the results were sorted in descending order using **sort -nr** to display the most frequent products at the top.

A terminal window with a dark purple background. The window title bar shows 'vboxuser@M' and three tabs: 'hduser@Mtech: ~', 'hduser@Mtech: ~/kafka', and 'hdus'. On the left, a vertical sidebar contains icons for a file manager, application launcher, and system settings. The terminal content shows a list of products: 'Shirt', 'Laptop', 'Mobile', 'Shoes', 'Laptop', 'Jacket', 'Watch', 'Laptop', 'Mobile', 'Shoes', 'Tablet', 'Laptop', 'Shirt'. Below this, three commands are executed: 1. 'cut -d " " -f2 purchases.txt | sort | uniq -c' resulting in a list of counts and product names. 2. 'cut -d " " -f2 purchases.txt | sort | uniq -c | sort' resulting in a sorted list of counts and product names. 3. 'cut -d " " -f2 purchases.txt | sort | uniq -c | sort' resulting in a list of counts and product names. The terminal ends with a prompt 'vboxuser@Mtech:~/purchase_analysis\$'.

```
Shirt
Laptop
Mobile
Shoes
Laptop
Jacket
Watch
Laptop
Mobile
Shoes
Tablet
Laptop
Shirt
vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt | sort | uniq -c
  3 Jacket
  2 Jeans
 14 Laptop
  8 Mobile
  5 Shirt
  9 Shoes
  4 Tablet
  5 Watch
vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt | sort | uniq -c | sort
 14 Laptop
  9 Shoes
  8 Mobile
  5 Watch
  5 Shirt
  4 Tablet
  3 Jacket
  2 Jeans
vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt | sort | uniq -c | sort
 14 Laptop
  9 Shoes
  8 Mobile
vboxuser@Mtech:~/purchase_analysis$
```

The **head -3** command was then used to display the top three most frequently purchased products.

Top 3 products:

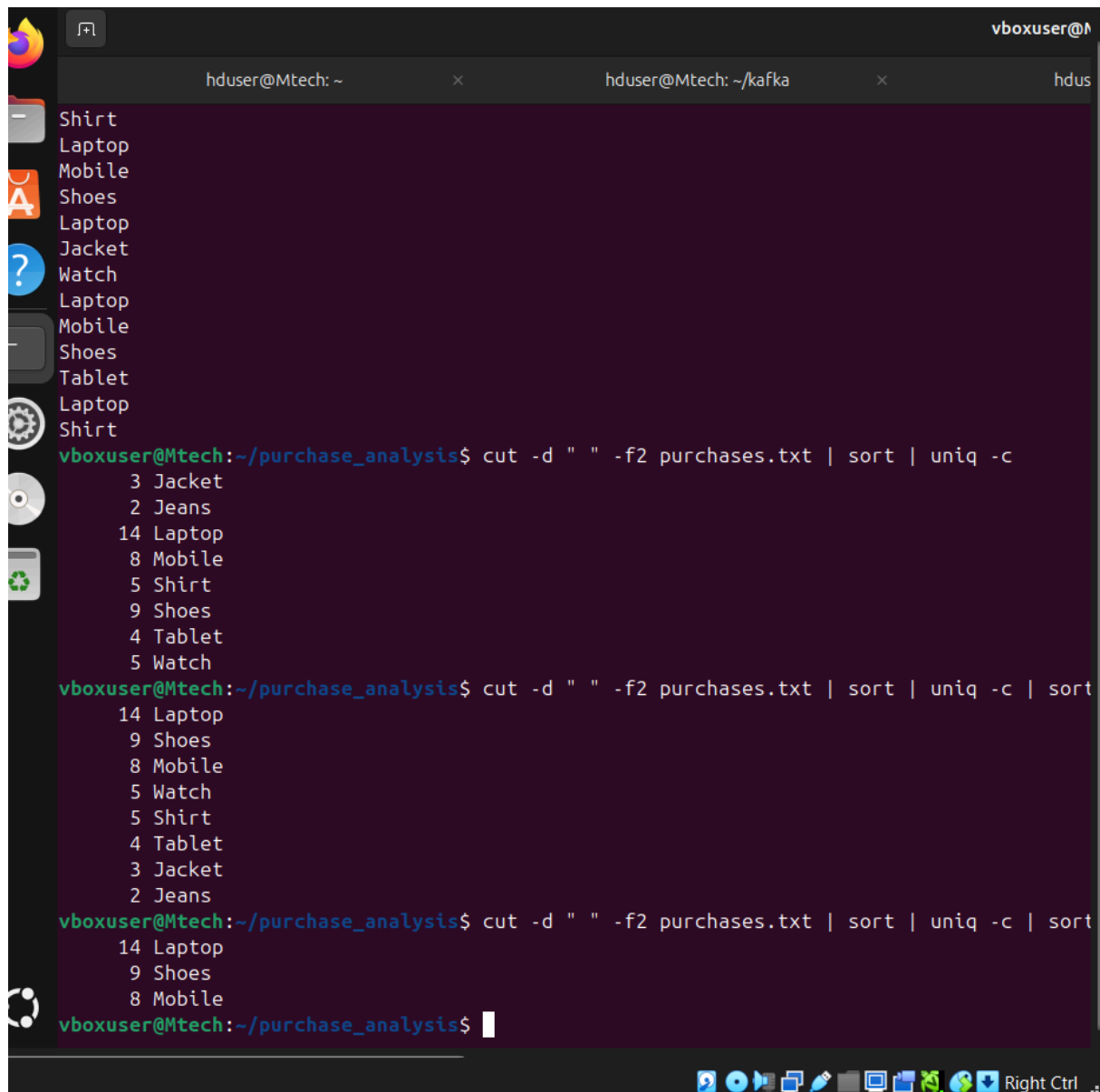
1. Laptop
2. Shoes
3. Mobile

This analysis helped identify the most popular products in the dataset.

5. Intermediate and Final Output

During the analysis, several intermediate outputs were generated such as sorted category lists, product frequency counts, and extracted columns. These outputs helped verify the correctness of each step before moving to the final analysis.

The final results indicated the total number of purchases, category distribution, and the most frequently purchased products.



```

vboxuser@Mtech: ~
x
hduser@Mtech: ~/kafka
x
hdus

Shirt
Laptop
Mobile
Shoes
Laptop
Jacket
Watch
Laptop
Mobile
Shoes
Tablet
Laptop
Shirt

vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt | sort | uniq -c
  3 Jacket
  2 Jeans
 14 Laptop
  8 Mobile
  5 Shirt
  9 Shoes
  4 Tablet
  5 Watch

vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt | sort | uniq -c | sort
 14 Laptop
  9 Shoes
  8 Mobile
  5 Watch
  5 Shirt
  4 Tablet
  3 Jacket
  2 Jeans

vboxuser@Mtech:~/purchase_analysis$ cut -d " " -f2 purchases.txt | sort | uniq -c | sort
 14 Laptop
  9 Shoes
  8 Mobile

vboxuser@Mtech:~/purchase_analysis$
```

6. Challenges Faced

During the activity, some challenges were encountered while understanding the correct usage of command pipelines. Combining multiple Unix commands using the pipe operator (|)

required careful attention to ensure that the output of one command was correctly passed as input to the next command.

Another challenge was understanding how sorting affects the behavior of the **uniq** command. It was observed that the **uniq** command only counts adjacent duplicate entries, so sorting the data before using **uniq** was necessary to obtain accurate results.

7. Conclusion

This activity demonstrated how Unix command-line tools can be effectively used for quick and efficient data analysis on structured text files. Tools such as **cut**, **sort**, **uniq**, **wc**, and **head** allow users to extract meaningful insights from datasets without requiring complex programming.

Through this analysis, it was possible to determine the total number of purchases, identify category-wise purchase distribution, and detect the most frequently purchased products. The activity also highlighted the importance of command pipelines in Unix for performing multi-step data processing tasks.

Overall, this practical exercise helped develop fundamental skills in text-based data analysis using Unix command-line tools, which are widely used in big data environments and real-world data processing workflows.